

Cell Structure and Function Lab Report

Objective

The objective of this laboratory exercise was to observe and identify the basic structures of plant and animal cells using light microscopy. Students learned to prepare wet mount slides and use proper staining techniques to enhance cellular structure visibility.

Materials and Methods

Materials used included:

- Light microscope (100x, 400x, 1000x magnification)
- Microscope slides and cover slips
- Methylene blue stain
- Iodine solution
- Elodea leaf samples
- Onion root tip samples
- Cheek cell samples (from student)

The procedure involved preparing wet mount slides for each sample type, applying appropriate stains, and observing under increasing magnification to identify cellular structures including the cell membrane, nucleus, cytoplasm, and in plant cells, the cell wall and chloroplasts.

Results

Question 1: Identify the structures you observed in the plant cells (Elodea).

In the Elodea leaf cells, I observed the following structures:

- **Cell Wall:** Visible as a thick outer boundary surrounding each cell
- **Cell Membrane:** Located just inside the cell wall
- **Chloroplasts:** Numerous green organelles distributed throughout the cytoplasm, showing movement (cyclosis)
- **Nucleus:** Visible as a darker, rounded structure when stained with iodine
- **Cytoplasm:** The gel-like substance filling the cell

Question 2: How do plant cells differ from animal cells based on your observations?

The main differences observed between plant and animal cells include:

- Plant cells have a rigid cell wall made of cellulose, while animal cells only have a flexible cell membrane
- Plant cells contain chloroplasts for photosynthesis, which were absent in the animal (cheek) cells
- Plant cells typically have larger, more centrally located vacuoles
- Animal cells are generally more irregular in shape, while plant cells maintain a more regular, rectangular shape due to the cell wall

Question 3: Explain the function of the nucleus based on what you observed.

The nucleus appeared as a darker-stained region within the cell, typically located near the center in plant cells and more

variable in position in animal cells. The nucleus functions as the control center of the cell, containing the cell's genetic material (DNA) and directing cellular activities including protein synthesis, cell division, and metabolism regulation.

Discussion

This laboratory exercise successfully demonstrated the fundamental differences and similarities between plant and animal cells. The observations confirmed the presence of membrane-bound organelles and highlighted how cellular structure relates to function. The use of appropriate staining techniques was crucial for visualizing certain structures, particularly the nucleus and cell membranes.

Potential sources of error included:

- Insufficient staining leading to poor contrast
- Air bubbles in wet mount preparations
- Using inappropriate magnification for structure identification

Conclusion

Through microscopic observation, we confirmed that both plant and animal cells share fundamental structures (nucleus, cell membrane, cytoplasm) while exhibiting key differences related to their specialized functions. Plant cells' additional structures like cell walls and chloroplasts support their role in photosynthesis and structural support.